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Software Engineering Experience

June 2025 – Present: Software Engineer Associate Director at HOWDEN RE (Global reinsurance broker)

Catalyst | Platform

[ReactJS](#) | [Vite](#) | [SCSS](#) | [OpenLayers](#) | [GeoPandas](#) | [Shapely](#) | [Axios](#) | [Python](#) | [FastAPI](#) | [Azure DevOps](#) | [Linear](#) | [Notion](#)

A multi-module automation and workflow platform built for Catastrophe Modellers to import, process, manage and visualise risk data, replacing significant reliance on external tooling across teams globally.

- Led the refactor of the Catalyst platform into ReactJS and FastAPI, integrating with the existing job management server and Azure architecture.
- Took ownership of the frontend across the full feature lifecycle, from requirements gathering and early design through to build, API integration, user support, demos and iterative improvement, growing the platform to a multi-module tool used daily by 30+ users globally.
- Identified the need for and began developing a map view and generation tool, to allow efficient visualisation of large location-based risk vector data, with functionality to generate brand-compliant map images for use in reports, to improve consistency and quality of outputs across teams.
- Designed and led a peer programming interview to identify and hire a new team member and managed their onboarding across the full frontend codebase and team processes.

Apr 2024 – June 2025: Software Engineer at Map Impact (Innovative Tech-for-Good Company)

Climate Risk Engine | Product

[Python](#) | [Poetry](#) | [Pandas](#) | [GeoPandas](#) | [Google Earth Engine](#) | [Jupyter Notebooks](#) | [GitHub](#)

- Transformed scientific methodology based in Jupyter Notebooks into a memory-optimised, Python engine to produce and validate our climate risk datasets, with detailed documentation.
- Built additional Python scripts for automating data manipulation, including a tool to allow internal users to extract samples from our climate risk datasets formatted for external presentations.

MapView Web Dashboard | External Tool

[ReactJS](#) | [Vite](#) | [SCSS](#) | [OpenLayers](#) | [proj4](#) | [turf/turf](#) | [Axios](#) | [GitHub](#)

- Led the design process of the dashboard, presenting wireframes and user stories to break down the functionality and describe how it should be implemented to the Engineering team.
- Developed the map-centric dashboard in ReactJS, managing API requests allowing users to create layers from our datasets using drawn or uploaded AOIs and wrote client-side functionality for visualising and manipulating the data on the map.

Environmental Monitoring of Welsh Agricultural Catchments | External Project

[ReactJS](#) | [Vite](#) | [SCSS](#) | [OpenLayers](#) | [proj4](#) | [PWAs](#) | [Docker](#) | [Helm](#) | [Kubernetes](#) | [GitHub](#)

- Designed, developed and presented to stakeholders, a map based dashboard prototype with OpenLayers in ReactJS, to help people in the agriculture community visualise and track data relating to water quality.

BiodiversityView and BiodiversityCheck | Product

[Python](#) | [Pandas](#) | [GeoPandas](#) | [Numpy](#) | [QGIS](#) | [Cartopy](#) | [GitHub](#) | [Jira](#) | [Confluence](#)

- Refactored the existing codebase, specifically removing the QGIS dependency and Python code improvements. Included running reports for stakeholder meetings and internal research and maintenance of configuration files and other assets across new versions of the code and methodology.

Other selected internal work and R&D

[Python](#) | [Poetry](#) | [Pandas](#) | [GeoPandas](#) | [Numpy](#) | [QGIS](#) | [Google Earth Engine](#) | [React](#) | [Vite](#) | [Axios](#) | [Typescript](#) | [ChartJS](#) | [TailwindCSS](#) | [SCSS](#) | [Rust](#) | [GeoServer](#) | [PostgreSQL](#) | [Swagger](#) | [Docker](#) | [Helm Charts](#) | [Kubernetes](#) | [GitHub](#) | [Jira](#)

- Developed a command line based QA tool in Python to produce validation reports in our BiodiversityView datasets generation pipeline and maintained this and our Topological Correction tool, including testing, implementing improvements and training team members how to use them.

- Created a simple web-hosted user interface in React to enable internal members to create samples from our datasets to use in investigations and presentations via our API.

Jan 2023 – Apr 2024: Software Developer at Spatial Days (Geospatial Tech Company)

City Explorer Toolkit - UK Centre for Ecology & Hydrology

[Django](#) (REST API) | [OAuth](#) | [ReactJS](#) | [NodeJS](#) | [SCSS](#) | [Swagger](#) (API docs) | [Trello](#) | [Notion](#) | [GitHub](#)

- Contributed to the design and development of the Toolkit's production version in React with Django REST API, including creating frontend pages and client-side functionality as well as refactoring the API.

Yemen Economic Tracking Initiative - ACAPS / Mercy Corps

[Python](#) | [Django](#) (REST API) | [PostgreSQL](#) (terminal + pgAdmin) | [HTML](#) | [CSS](#) | [Bootstrap](#) | [JS](#) | [jQuery](#) | [Slack](#) | [GitHub](#)

- Developed frontend and backend functionality using the Django framework and contributed to designs that provide data visualisation on economic trends and policy for Yemen.

Spatio-Temporal Asset Catalogue (STAC) Portal - Ordnance Survey

[Django](#) | [FastAPI](#) | [Python](#) | [ReactJS](#) | [SCSS](#) | [STAC Standard](#) | [Rasterio](#) | [Trello](#) | [Hoppscotch](#) | [Swagger](#) (API docs) | [GitHub](#)

- Designed, created and refactored a number of pages and features in ReactJS for a STAC management platform to streamline the Ordnance Survey's handling of aerial and satellite imagery data.
- Assisted in the portal's API design in Python with FastAPI, as well as developing two Python microservices for validating OS-specific imagery at different points of the processing pipeline.

Other selected work

[WordPress](#) | [Liquid](#) | [SSL Certificates](#) | [Google Calendar API](#) | [Python](#) | [Snapista](#) | [Rasterio](#) | [GDAL](#) | [PyProj](#) | [ESA SNAP](#) | [QGIS](#) | [GeoServer](#) | [PostgreSQL](#) | [Docker](#) | [Helm](#) | [Kubernetes](#) | [S3 Proxy](#) | [Azure](#) | [Postman](#) | [Notion](#) | [Trello](#) | [GitHub](#)

- Refactored and rebuilt a GeoServer-based Web Map Service for OpenStreetMap data, implementing modern Python techniques, GeoServer REST API, Postgres database configured with PostGIS and aided in the redeployment of the project in Azure with Kubernetes and Helm Charts.
- For the [JPP CommonSensing Project](#) with Satellite Applications Catapult: Identified and resolved issues in existing Synthetic Aperture Radar (SAR) processing pipelines using Python and ESA's SNAP GPT, completed bug fixing and testing in the frontend application and assisted in the Dockerization of the project and configuration of GitHub actions and Azure pipelines.

Other Selected Experience

Nov 2021 – Dec 2022: Kitchen Member (4mo) Kitchen Team Leader (9mo) - Toby Carvery

Sep 2020 – Aug 2021: Police Constable - Metropolitan Police Service

Oct 2019 – Aug 2020: Quality and Compliance Officer / Team Leader - DHL

Feb 2018 – May 2018: University of Hertfordshire – Python Programmer (contract) [Python](#) | [MS Excel](#)

Education

Sep 2021 – Sep 2022: Code Institute – Diploma in Full Stack Software Development

[Django](#) | [Python](#) | [Flask](#) | [JavaScript](#) | [HTML](#) | [CSS](#) | [Google Sheets API](#) | [PostgreSQL](#) | [GitHub](#) | [Heroku](#) | [Slack](#) | [Balsamiq](#)

[Midi Dragon](#) – Music Products and Services

[Cook eBook](#) – Recipe Making Community

[Blosh](#) – Blog Website for a Student

[Python Minigames](#) – For the Python Console

[Fretboard Trainer](#) – Multichoice Quiz Game

[Rock Climbing](#) – Informational Website

Sep 2016 – Sep 2019: University of Hertfordshire – BSc (Hons) Physics (1st Class)

[Awards: University Prize for Outstanding Performance on the Physics Program](#)

- **Dissertation** – Investigating Cloud-Cluster Collisions: Can a Molecular Cloud Capture Stars from a Star Cluster During a Cloud-Cluster Collision?
- **Selected modules** – Applications of Computing (MATLAB) | Programming (Python) | (Partial) Differential Equations | Computational Physics (MATLAB / Python) | Contemporary Quantum Physics (Maple)

Other Interests

Music (guitar, piano, drums and [production](#)) | Rock Climbing | Hiking | [Video/Photo Editing](#) | Reading | Travel